Prayas JEE AIR (2025)

WEEKLY TEST 23

DURATION : 180 Minutes | **DATE : 22/12/2024** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25 questions** in each part in which first **20** questions are of Objective Type and Last **5 questions** are integers type and all **25 questions are compulsory**.
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number: _____

OMR Bar Code Number: _____

Candidate's Signature: _____ **Invigilator's Signature** _____

SECTION-I (PHYSICS)

Single Correct Type Questions

1. When a capillary is dipped vertically inside a liquid, the liquid rises up to a height h. In the same capillary, the same liquid is filled in horizontal position and then kept in vertical position with both ends in air. The maximum length of liquid column which can be kept in vertical position is

(1) $\frac{h}{2}$ (2) h

(3) $2h$ (4) $\frac{3h}{2}$

2. A long solenoid of radius R carries a time(t) dependent current $I(t) = I_0 t(\text{l}-t)$. A ring of radius $2R$ is placed coaxially near its middle. During the time interval $0 \le t \le 1$, the induced current (I_R) and the induced emf (V_R) in the ring change as:

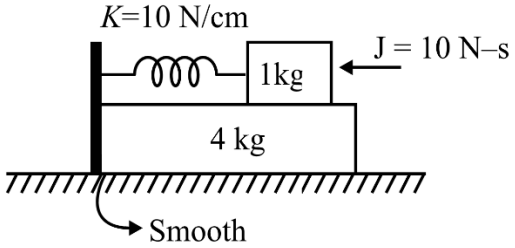
(1) Direction of I_R remains unchanged and V_R is zero at $t = 0.25$

(2) At $t = 0.5$ direction of I_R reverses and V_R is zero

(3) At $t = 0.25$ direction of I_R reverses and V_R is maximum

(4) Direction of I_R remains unchanged and V_R is maximum at $t = 0.25$

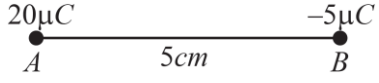
3. A smooth plank of mass 4 kg holds a block of mass 1 kg which is attached to this plank with the help of a massless spring as shown. If a sharp impulse J = 10 N-s is given to the block, then maximum compression of the spring will be



(1) $\sqrt{2}$ m (2) $\frac{\sqrt{2}}{5}$ m

(3) $\frac{1}{5}$ m (4) $\frac{\sqrt{2}}{50}$ m

4. Two particles A and B having charges 20 μC and –5 μC respectively are held fixed with a separation of 5 cm. At what position a third charged particle should be placed so that it does not experience a net electric force?



(1) At 5 cm from 20 μC on the left side of system.

(2) At 1.25 cm from –5 μC between two charges.

(3) At midpoint between two charges.

(4) At 5 cm from –5 μC on the right side.

5. A series L-R circuit is connected to a battery of emf V. If the circuit is switched on at $t = 0$, then the time at which the energy stored in the inductor reaches $\left(\frac{1}{n}\right)$ times of its maximum value, is

(1) $\frac{L}{R}\ln\left(\frac{\sqrt{n}+1}{\sqrt{n}-1}\right)$ (2) $\frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}-1}\right)$

(3) $\frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}+1}\right)$ (4) $\frac{L}{R}\ln\left(\frac{\sqrt{n}-1}{\sqrt{n}}\right)$

6. In a typical combustion engine the work done by a gas molecule is given by $W = \alpha^2\beta e^{-\frac{\beta x^2}{kT}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature. If α and β are constants, dimensions of α will be:

(1) $\left[MLT^{-1}\right]$

(2) $\left[M^0LT^0\right]$

(3) $\left[M^2LT^{-2}\right]$

(4) $\left[MLT^{-2}\right]$

7. The electric field of a plane electromagnetic wave is given by

$$\vec{E} = E_0\frac{\hat{i}+\hat{j}}{\sqrt{2}}\cos(kz+\omega t)$$

At t =0, a positively charged particle is at the point $(x, y, z) = \left(0, 0, \frac{\pi}{k}\right)$. If its instantaneous velocity at (t = 0) is $v_0\hat{k}$, force acting on it due to the wave is:

(1) Zero

(2) Parallel to $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

(3) Parallel to $\hat{k}$

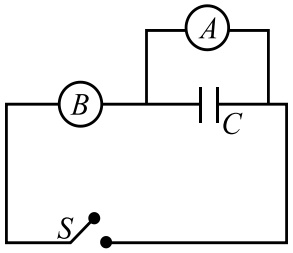
(4) antiparallel to $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

8. Match **Column**–I with **Column**–II
Choose the correct answer from the options given below:

	Column-I		**Column-II**
A	Moment of inertia of solid sphere of radius R about any tangent.	P	$\frac{5}{3}MR^2$
B	Moment of inertia of hollow sphere of radius (R) about any tangent	Q	$\frac{7}{5}MR^2$
C	Moment of inertia of circular ring of radius (R) about its diameter.	R	$\frac{1}{4}MR^2$
D	Moment of inertia of circular disc of radius (R) about any diameter.	S	$\frac{1}{2}MR^2$

(1) A→Q; B→P; C→S; D→R
(2) A→P; B→Q; C→S; D→R
(3) A→Q; B→P; C→R; D→S
(4) A→P; B→Q; C→R; D→S

9. A capacitor of capacitance *C* is connected to two voltmeters (*A* and *B*) as shown in figure. Voltmeter *A* is ideal, having infinite resistance, while voltmeter *B* has some finite resistance. Initially the capacitor is charged and then the switch (*S*) is closed. The reading of *A* and *B* will be equal



(1) Only after long time
(2) After time $t = RC$
(3) After time $t = RC(\ln 2)$
(4) At all times

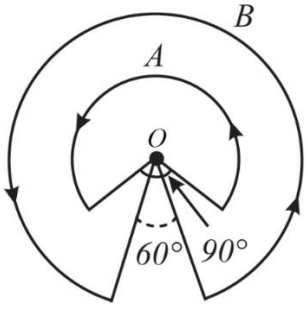
10. At time $t = 0$ a particle starts travelling from a height $7\,\hat{z}$ in a plane keeping z coordinate constant. At any instant of time it's position along the $\hat{x}$ and $\hat{y}$ directions are defined as $3t$ and $5t^3$ respectively. At $t = 1$s acceleration of the particle will be

(1) $-30\hat{y}$ (2) $30\hat{y}$
(3) $3\hat{x}+15\hat{y}$ (4) $3\hat{x}+15\hat{y}+7\hat{z}$

11. Given below are two statements:
Statement-I: An electric dipole is placed at the center of a hollow sphere. The flux of electric field through the sphere is zero but the electric field is not zero anywhere in the sphere.
Statement-II: If *R* is the radius of a solid metallic sphere and *Q* be the total charge on it. The electric field at any point on the spherical surface of radius $r(<R)$ is zero but the electric flux passing through this closed spherical surface of radius *r* is not zero.
In the light of the above statements. Choose the correct answer from the option given below:
(1) Statement-I is true but Statement-II is false
(2) Statement-I is false but Statement-II is true
(3) Both Statement-I and Statement-II are true
(4) Both Statement-I and Statement-II are false

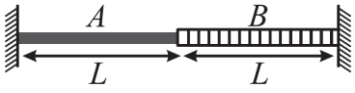
12. Ship *A* is sailing towards north-east with velocity $= 30\,\hat{i} + 50\,\hat{j}$ km/hr where $\hat{i}$ points east and $\hat{j}$, north. Ship *B* is at a distance of 80 km east and 150 km north of Ship *A* and is sailing towards west at 10 km/hr. *A* will be at minimum distance from *B* in:
(1) 1.3 hrs. (2) 3.1 hrs.
(3) 4.2 hrs. (4) 2.6 hrs.

13. A wire *A*, bent in the shape of an arc of a circle, carrying a current of 2 A and having radius 2 cm and another wire *B*, also bent in the shape of arc of a circle, carrying a current of 3 A and having radius of 4 cm, are placed as shown in the figure. The ratio of the magnetic fields due to the wires *A* and *B* at the common centre *O* is



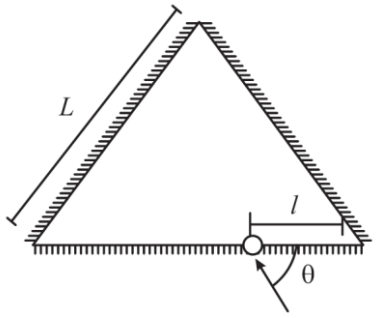
(1) 2 : 5 (2) 6 : 5
(3) 4 : 5 (4) 6 : 4

14. A wire of length $2L$, is made by joining two wires *A* and *B* of same lengths but different radii *r* and $2r$ and made of the same material. It is vibrating at a frequency such that the joint of the two wires forms a node. If the number of anti-nodes in wire *A* is *p* and that in *B* is *q* then the ratio $p : q$ is:



(1) 4 : 9
(2) 3 : 5
(3) 1 : 4
(4) 1 : 2

15. Three plane mirrors form an equilateral triangle with each side of length L. There is a small hole at distance $l > 0$ from one of the corners as shown in the figure. A ray of light is passed through the hole at an angle θ and can only come out through the same hole. The cross section of the mirror configuration and the ray of light lie on the same plane.



Which of the following statement is correct?

(1) The ray of light will never come out for θ = 30°, for $0 < l < \frac{L}{2}$.

(2) There is an angle for $l = \frac{L}{2}$ at which the ray of light will come out after two reflections.

(3) The ray of light will never come out for θ = 60°, and $l = \frac{L}{3}$

(4) The ray of light will come out for θ = 60°, and $0 < l < \frac{L}{2}$ after six reflections.

16. A particle of mass m is moving with speed $2v$ collides with a mass $2m$ moving with speed v in the same direction. After collision, the first mass is stopped completely while the second one splits into two particles each of mass m, which move at angle 45° with respect to the original direction. The speed of each of the moving particle will be:

(1) $v/(2\sqrt{2})$ (2) $2\sqrt{2}v$

(3) $\sqrt{2}v$ (4) $v/\sqrt{2}$

17. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and λ/2 if the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface of the material is :

(h = Planck's constant, c = speed of light)

(1) $\frac{2hc}{\lambda}$ (2) $\frac{hc}{3\lambda}$

(3) $\frac{hc}{2\lambda}$ (4) $\frac{hc}{\lambda}$

18. An ideal gas has an adiabatic exponent r. In some process its molar heat capacity varies as $C = \frac{\alpha}{T}$, where α is a constant. The work done by one mole of the gas during its heating from the temperature T_0 to the temperature ηT_0

(1) $\alpha \ln\eta - \frac{RT_0(\eta-1)}{r-1}$

(2) $\alpha \ln\frac{1}{\eta} - \frac{RT_0(\eta-1)}{r-1}$

(3) $\alpha \ln\eta + \frac{RT_0(\eta-1)}{r-1}$

(4) $\alpha \ln\frac{1}{\eta} + \frac{RT_0(\eta-1)}{r-1}$

19. The nucleus of an atom decays by emitting an α-particle as per the equation

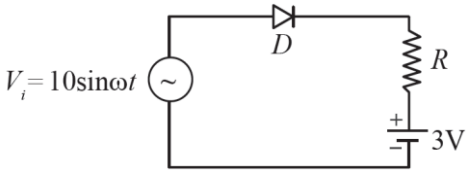
$^{235}_{92}Y \longrightarrow {}^{231}_{90}X + {}^{4}_{2}He + Q$

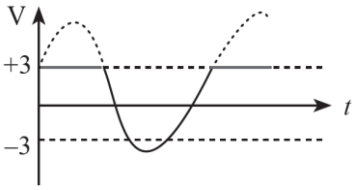
Binding energy per nucleon of Y, X, α are 7.8 MeV, 7.835 MeV, 7.07 MeV, respectively. Find Q-value in MeV.

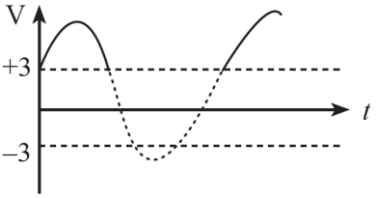
(1) 4.78 (2) 5.16

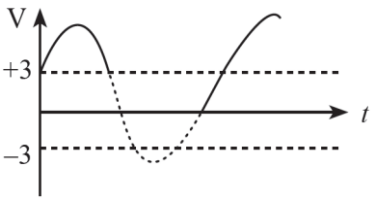
(3) 4.37 (4) 5.78

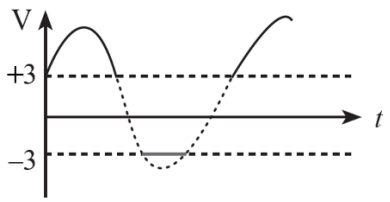
20. Choose the correct waveform that can represents the voltage across R of the following circuit, assuming the diode is ideal one:



(1)


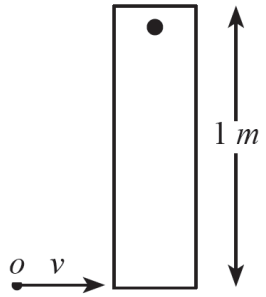
(2)


(3)


(4)


Integer Type Questions

21. A thin rod of mass 0.9 kg and length 1 m is suspended, at rest, from one end so that it can freely oscillate in the vertical plane. A particle of mass 0.1 kg moving in a straight line with velocity 80 m/s hits the rod at its bottom most point and sticks to it (see figure). The angular speed (in rad/s) of the rod immediately after the collision will be



22. A fish rising vertically upward with a uniform velocity of 8 ms^{-1}, observes that a bird is diving vertically downward towards the fish with the velocity of 12 ms^{-1}. If the refractive index of water is 4/3, then the actual velocity of the diving bird to pick the fish, will be _____ ms^{-1}.

23. First, a set of *n* equal resistors of 10 Ω each are connected in series to a battery of emf 20V and internal resistance 10 Ω. A current *I* is observed to flow. Then, the *n* resistors are connected in parallel to the same battery. It is observed that the current is increased 20 times, then the value of *n* is ____.

24. In a double slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-3} cm. If the distance between the slits is 1 mm, then the wavelength of the light will be _____nm.

25. In a screw gauge, there are 100 divisions on the circular scale and the main scale moves by 0.5 mm on a complete rotation of the circular scale. The zero of circular scale lies 6 divisions below the line of graduation when two studs are brought in contact with each other. When a wire is placed between the studs, 4 linear scale divisions are clearly visible while 46th division the circular scale coincide with the reference line. The diameter of the wire is _____ $\times 10^{-2}$ mm.

SECTION-II (CHEMISTRY)

Single correct Type Questions

26. The bond order and the magnetic characteristics of CN^- are

(1) $2\frac{1}{2}$, paramagnetic (2) 3, diamagnetic

(3) $2\frac{1}{2}$, diamagnetic (4) 3, paramagnetic

27. Consider the following balanced redox reaction:

$aFe^{2+}(aq) + bCr_2O_7^{2-}(aq) + cH^+(aq) \rightarrow xFe^{3+}(aq) + yCr^{3+}(aq) + zH_2O(l)$

Select the correct relation.

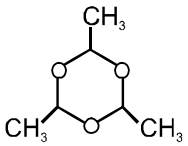
(1) a + b + c = x + y + z always
(2) a + b = x + y always
(3) a = x, b = 2y and c = 7z
(4) a = x, z = 7b and c = 2z

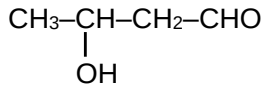
28. Consider the reaction

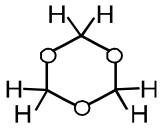
$3CH_3CHO \xrightarrow[298\,K]{H^{\oplus}} P.$

It is given that the product P is a pleasant smelling liquid with a boiling point of 397 K. Also given is the fact that P is used as a hypnotic. The correct structure representing 'P' is:

(1) $CH_3 - CH = CH - CHO$

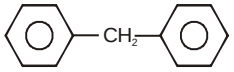
(2) 

(3) 

(4) 

29. Among the given options, identify the option representing incorrect statement?

(1) Ammonium salts on reaction with strong alkalies produce a non-inflammable gas.
(2) fac-$[Co(NO_3)_3(dien)]$ is achiral.
(3) Solid potassium dichromate and concentrated H_2SO_4 liberates chlorine gas with solid sodium chloride which is used to confirm presence of $Cl^{\ominus}$ ion.
(4) The increasing order of paramagnetic properties amongst VCl_3, $VOSO_4$, Na_3VO_4, $[V(H_2O)_6]\ SO_4.H_2O$ is $Na_3VO_4 < VOSO_4 < VCl_3 < [V(H_2O)_6]\ SO_4.H_2O$.

30. Diphenyl methane (), has a molecular formula $C_{13}H_{12}$. When one of the hydrogen atom is replaced by chlorine atom, calculate the number of structural isomers possible.

(1) 4
(2) 8
(3) 7
(4) 6

31. Consider the following statements:

Statement 1: The acidic strength of HBr is greater than HI due to hydrogen bonding.

Statement 2: Hydration energy of $F^{\ominus}$ ion is greater than $Cl^{\ominus}$ ion.

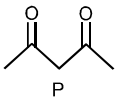
Statement 3: Because of three 'O' atoms, O_3 is more powerful oxidising agent when compared with F_2.

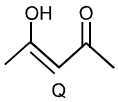
Statement 4: Polyhalides are not formed by fluorine.

The correct arrangement of above statements in terms of true/false is

(1) T T T F
(2) F T F T
(3) T F T F
(4) F F T T

32. Identify the option representing **incorrect** statement regarding P and Q?


P


Q

(1) P & Q are tautomers
(2) P & Q are positional isomers
(3) Q is a stable enol
(4) Q has resonance stability

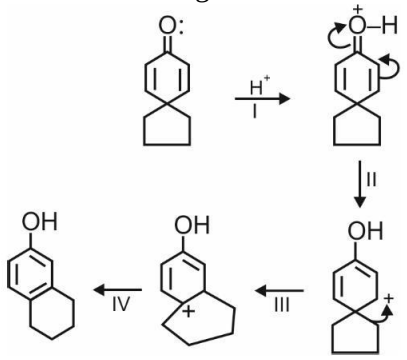
33. $(X) + K_2CO_3 \xrightarrow{\text{heat}} (Y)$

$(Y) + Cl_2 \longrightarrow (Z)\,\text{Pink}$

Which of the following is correct?

(1) X = black, MnO_2, Y = Blue, K_2CrO_4, Z = $KMnO_4$
(2) X = green, Cr_2O_3, Y = yellow, K_2CrO_4, Z = $K_2Cr_2O_7$
(3) X = black, MnO_2, Y = green, K_2MnO_4, Z = $KMnO_4$
(4) X = black, Bi_2O_3, Y = colourless, $KBiO_2$, Z = $KBiO_3$

34. Consider the following mechanism.



Identify the correct statement.

(1) Step-II is the RDS of the reaction
(2) Step-III is the RDS of the reaction
(3) Overall ΔH of the reaction is positive
(4) In step-III, the degree of unsaturation is increased by one

35. **Statement-1:** In acidic medium Zn^{2+} is not precipitated by S^{-2} ions.

and

Statement-2: Common ion effect increases the concentration of S^{-2} ions

(1) Statement-1 is True, statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(3) Statement-1 is True, Statement -2 is False.
(4) Statement-1 is False, Statement-2 is True.

36. Consider the following statements.

A – Magnetic quantum no. gives information about spatial orientation of orbitals

B – Energy of an orbital depends on principal quantum no. for unielectronic species.

C – Spin quantum no. is also obtained from Schrodinger wave equation

The correct statement(s) is/are

(1) Only A
(2) Only A and B
(3) Only A and C
(4) A, B and C

37. Stearic acid [$CH_3(CH_2)_{16}CO_2H$] is a fatty acid, the part of fat that stores most of the energy. 1.0 g of stearic acid was burned in a bomb calorimeter. The bomb had a heat capacity of 652 J/°C. If the temperature of 500 g water (Specific heat capacity of water = 4.18J/g°C) rose from 25.0 to 39.3°C, how much heat was released when the stearic acid was burned?

(1) 39.21 kJ
(2) 29.91 kJ
(3) 108 kJ
(4) 9.32 kJ

38. A mixture of A and B in the molar ratio 1:2 forms a maximum boiling azeotrope. Identify the incorrect statement, if A is more volatile [Molar mass of A=100, Molar mass of B = 50]

(1) A liquid solution of A & B having mass % of A=50 will have vapours having mass % of A =50.

(2) A liquid solution A and B having mass % of A > 50% will have vapours having mass % of A > 50%.

(3) A mixture of A & B in the molar ratio 1:3 can be separated into azetropic mixture and pure A.

(4) A mixture of A and B in the molar ratio 2:3 can be separated into azeotropic mixture and pure A.

39. An optically inactive amine (A) is methylated with excess of CH_3I and there after reacted with AgOH to form an alkene (B). The alkene (B) after ozonolysis gives HCHO and $CH_3CH_2CH_2CHO$. (A) and (B) in the reaction are, respectively,

(1) $CH_3–CH_2–CH_2–NH_2$ & $CH_3–CH_2–CH = CH_2$

(2) $CH_3–CH_2–\underset{\displaystyle NH_2}{\underset{|}{CH}}–CH_2 – CH_3$ &
$CH_3–CH_2–CH = CH–CH_3$

(3) $CH_3–CH_2–CH_2–CH_2–CH_2–NH_2$ &
$CH_3–CH_2–CH_2–CH = CH_2$

(4) $CH_3–CH_2–CH_2–\underset{\displaystyle NH_2}{\underset{|}{CH}}–CH_3$ &
$CH_3–CH_2–CH_2–CH = CH_2$

40. (A) $M_g^- \rightarrow M_{(g)}$ (B) $M_{(g)} \rightarrow M_{(g)}^+$

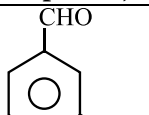
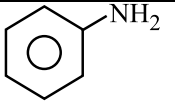
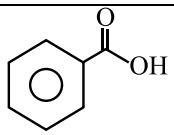
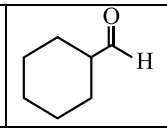
(C) $M_{(g)}^+ \rightarrow M_{(g)}^{2+}$ (D) $M_{(g)}^{2+} \rightarrow M_{(g)}^{3+}$

Minimum and maximum I.E. would be of:

(1) a,d (2) b,c

(3) c,d (4) d,a

41. Match the following

Column-I (Compound)		Column-II (Identifying test)	
(A)		(P)	Carbylamine test
(B)		(Q)	Fehling's solution test
(C)		(R)	$NaHCO_3$ test
(D)		(S)	Neutral $FeCl_3$ test

(1) A→S; B→P; C→Q; D→R

(2) A→S; B→P; C→R; D→Q

(3) A→Q; B→S; C→R; D→P

(4) A→Q; B→P; C→R; D→S

42. Consider the following sequence of reactions:

$$CH_3CH_2CH_2C\equiv CH \xrightarrow[H^+]{H_2O} (A) \xrightarrow{NaBH_4} (B) \xrightarrow{SOCl_2} (C) \xrightarrow{KCN(aq)} (D) \xrightarrow{H_2O/H^+} (E) \xrightarrow[\text{Red P}]{Br_2} (F)$$

Correct structure of product 'F' is

(1) $CH_3CH_2CH_2\underset{\displaystyle COOH}{\underset{|}{CH}}–CH_3$

(2) $CH_3CH_2CH_2\underset{\displaystyle Br}{\underset{|}{CH}}COOH$

(3) $CH_3CH_3CH_2–\underset{\displaystyle Br}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}–COOH$

(4) $CH_3CH_3CH_2\underset{\displaystyle COOH}{\underset{|}{CH}}–CH_2Br$

43. What would be the reduction potential of an electrode at 298 K, which originally contained 1 M $K_2Cr_2O_7$ solution in acidic buffer solution of pH = 1.0 and which was treated with 50% of the Sn necessary to reduce all $Cr_2O_7^{2-}$ to Cr^{3+}. Assume pH of solution remains constant.

Given: $E^\circ_{Cr_2O_7^{2-}/Cr^{3+},H^+} = 1.33V; \log 2 = 0.3;$

$\frac{2.303RT}{F} = 0.06$

(1) 1.285 V

(2) 1.193 V

(3) 1.187 V

(4) 1.473 V

44. Consider the following reaction sequence

$$PhCOOH \xrightarrow{NH_3} [A] \xrightarrow{Heat} [B] \xrightarrow{P_2O_5} [C] \xrightarrow[Ni/\Delta]{H_2} [D]$$

$$[B] \xrightarrow[Heat]{Br_2/KOH} [E]$$

Which of the following is false?

(1) [A] is $PhCONH_2$

(2) [C] is dehydration product of [B]

(3) [B] to [E] is hoffmann bromamide rearrangement

(4) [D] and [E] are homologous

45. Given below are two statements:
One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The covalent character of $FeCl_3$ is more than that of $FeCl_2$.

Reason (R): The covalent character of the compound increases with increase in the charge of the cation

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(3) (A) is correct but (R) is incorrect.
(4) (A) is incorrect but (R) is correct.

Integer type Questions

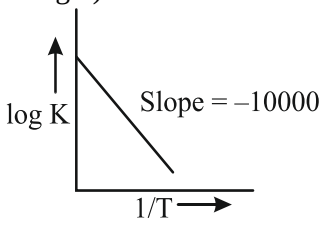
46. A metal oxide has the formula M_2O_3. This oxide reacts with hydrogen to produce M and H_2O according to the reaction (unbalanced) $M_2O_3 + H_2 \rightarrow M + H_2O$. If 32 g of M_2O_3 required 1.2 g of hydrogen for complete reduction, the atomic mass of metal M is (in gm/mol)

47. In the cobalt-carbonyl complex $[Co_2(CO)_8]$, number of Co-Co bonds is "X" and number of terminal CO ligands is "Y". The value of X + Y is _____.

48. Consider the following polypeptide.
Gly – Ala – Val – Gly – Leu – Ile – Asp – Gly
After hydrolysis the number of different optically active amino acids obtained are x and different optically inactive amino acids are y then find $\frac{x}{y}$:

49. Equilibrium constants for ionization of T_2O and H_2O are different at 298 K. Pure T_2O has pT (like pH) equal to 7.62, then pT of solution prepared by adding 10 ml of 0.2 M TCl to 15 ml of 0.25 M NaOT is [Given: log7 = 0.845] (Nearest integer)

50. Given below is a plot of log k vs $\frac{1}{T}$ for a reaction
$aA + bB \rightarrow cC + dD$, slope = –10000.
If the rate constant of the reaction is 10^{-5} s^{-1} at 500 K, then the temperature at which the rate constant of the reaction is 10^{-4} s^{-1} is___K.
(Nearest integer)



SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. If $A=\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$, then $(I+A)^4$ equals

(1) $I + 15A$ (2) $I + 7A$
(3) $I + 8A$ (4) $I + 11A$

52. The expression
$$\left[\left(\frac{2x}{1+\sqrt{4x+1}}\right)^7-\left(\frac{2x}{1-\sqrt{4x+1}}\right)^7\right]\frac{1}{\sqrt{4x+1}}$$
is a polynomial in x of degree
(1) 4 (2) 6
(3) 7 (4) 3

53. If α, β are the roots of $4x^2 - 15x + \lambda = 0$, where $\lambda \in R$, such that $1 < \alpha < 2$ and $2 < \beta < 3$, then the sum of integral values of λ is
(1) 16
(2) 61
(3) 25
(4) 52

54. The solution of the differential equation
$\frac{dy}{dx}=\frac{x^2+y^2-4x-4y+8}{xy-2x-2y+4}$ is
(1) $(x-y)^2=\ln(x-1)+c$
(2) $\frac{(y-2)^2}{(x-2)^2}=2\ln|x-2|+c$
(3) $(x-y)^2=\ln(x-y)+c$
(4) $(x-2)^2=2\ln|x-2|+c(y-2)^2$

55. Sides of a triangle formed by the lines $(x + y - 1)(x^2 - xy) = 0$ then match the centre of triangle in **Column**-I to the coordinates in **Column**-II

	Column-I		**Column-II**
A	Orthocentre	P	(1/6, 1/2)
B	Circumcentre	Q	$(1/(2+2\sqrt{2}), 1/2)$
C	Centroid	R	(0, 1/2)
D	Incentre	S	(1/2, 1/2)

(1) A→S; B→R; C→P; D→Q
(2) A→P; B→Q; C→R; D→S
(3) A→S; B→P; C→R; D→Q
(4) A→R; B→S; C→P; D→Q

56. The shortest distance between the line joining the points (1, 2, 3) and (2, 3, 4), and the line $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-2}{0}$ is d, then $14d^2$ equal to
(1) 9
(2) 16
(3) 25
(4) 4

57. A circle of constant radius 'a' passes through origin 'O' and cuts the axes of co-ordinates in points P and Q, then the equation of the locus of the foot of perpendicular from O to PQ is

(1) $\left(x^2+y^2\right)\left(\frac{1}{x^2}+\frac{1}{y^2}\right)=4a^2$

(2) $\left(x^2+y^2\right)^2\left(\frac{1}{x^2}+\frac{1}{y^2}\right)=a^2$

(3) $\left(x^2+y^2\right)^2\left(\frac{1}{x^2}+\frac{1}{y^2}\right)=4a^2$

(4) $\left(x^2+y^2\right)\left(\frac{1}{x^2}+\frac{1}{y^2}\right)=a^2$

58. If the complex number $z = x + iy$ moves in argand plane such that $1 \le |x + iy| \le 2$ and $y \ge x$ (where $x, y \in$ R), determines a region with area k, then value of the definite integral $\int_0^{2k/3} \{\sin x\}\,dx$ equal to (where $\{k\}$ represents fractional part of k)

(1) 0
(2) 1
(3) 2
(4) 3

59. 100 identical objects are distributed among 10 persons in which 1st person gets at least one object, 2nd person gets at least 2 objects, 3rd person gets at least 3 objects, …, 10th person gets at least 10 objects, then total number of ways of distribution is

(1) ${}^{55}C_{10}$ (2) ${}^{55}C_9$
(3) ${}^{54}C_9$ (4) ${}^{54}C_{44}$

60. If $\sum_{n=1}^{\infty}\tan^{-1}\left(\frac{\sqrt{2n+2}-\sqrt{2n-2}}{2+\sqrt{n^2-1}}\right)=\frac{\pi}{2}+\tan^{-1}k$, then find the value of k^2 is

(1) $\sqrt{2}$
(2) 3
(3) 1
(4) 2

61. The ellipse $4x^2 + 9y^2 = 36$ and the hyperbola $a^2 x^2 - y^2 = 4$ intersects at right angles then the equation of the circle through the points of intersection of two conic is

(1) $x^2 + y^2 = 5$

(2) $\sqrt{5}\left(x^2+y^2\right)-3x-4y=0$

(3) $\sqrt{5}\left(x^2+y^2\right)+3x+4y=0$

(4) $x^2 + y^2 = 25$

62. The minimum value of the function $f(x)=\int_0^x \frac{d\theta}{\cos\theta}+\int_x^{\pi/2}\frac{d\theta}{\sin\theta}$ where $x \in \left(0,\frac{\pi}{2}\right)$, is

(1) $2\ln(\sqrt{2}+1)$ (2) $\ln(2\sqrt{2}+2)$
(3) $\ln(2+\sqrt{3})$ (4) $\ln(3+\sqrt{2})$

63. Let $\lambda \in R, \vec{a}=\lambda\hat{i}+2\hat{j}-3\hat{k}, \vec{b}=\hat{i}-\lambda\hat{j}+2\hat{k}$.
If $\left(\left(\vec{b}-\vec{a}\right)\cdot\left(\vec{a}\times\vec{b}\right)\right)\left(\vec{a}+\vec{b}\right)-\left(\left(\vec{b}-\vec{a}\right)\cdot\left(\vec{a}+\vec{b}\right)\right)\left(\vec{a}\times\vec{b}\right)$
$=8\hat{i}-40\hat{j}-24\hat{k}$, then $\left|\lambda\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}-\vec{b}\right)\right|^2$ is equal to

(1) 140 (2) 132
(3) 144 (4) 136

64. If α, $\beta(\alpha < \beta)$ are the roots of the equation $g(x) = ax^2 + bx + c = 0$ and $f(x)$ be an even function, then $\int_\alpha^\beta \frac{e^{f\left(\frac{g(x)}{x-\alpha}\right)}}{e^{f\left(\frac{g(x)}{x-\alpha}\right)}+e^{f\left(\frac{g(x)}{x-\beta}\right)}}dx$ is equal to

(1) $\frac{1}{2}(\beta-\alpha)$ (2) $\frac{1}{2}(\beta^2-\alpha^2)$

(3) $\frac{1}{2}(\beta+\alpha)$ (4) 0

65. For real valued functions $f_i(x) = \sin 2p_i x$, $\forall\ i = 1, 2, 3$; $p_i \in N$ such that period of $f_1(x) - f_2(x) + 3f_3(x)$ is π, period of $f_1(x) + 2f_2(x)$ is $\frac{\pi}{2}$ and period of $f_1(x) + 10f_3(x)$ is $\frac{\pi}{3}$ then min $(p_1 + p_2 + p_3)$ is

(1) 5 (2) 10
(3) 11 (4) 8

66. Let S be the focus of the parabola $y^2 = 8x$ and PQ be the common chord of the circle $x^2 + y^2 - 2x - 4y = 0$ and the given parabola. The area of the triangle PQS is

(1) 2 (2) 3
(3) 5 (4) 4

67. $\int \tan x \cdot \tan 2x \cdot \tan 3x\ dx =$

(1) $-\ell n|\cos x|-\frac{1}{2}\ell n|\sec 2x|+\frac{1}{3}\ell n|\sec 3x|+C$

(2) $-\ell n|\sec x|-\frac{1}{2}\ell n|\sec 2x|+\frac{1}{3}\ell n|\sec 3x|+C$

(3) $\ell n|\cos x|+\ell n|\cos 2x|+\ell n|\cos 3x|+C$

(4) $\ell n|\sec x|+\frac{1}{2}\ell n|\sec 2x|+\frac{1}{3}|\sec 3x|+C$

68. The coefficient of variation of numbers 1,2,3,.....,n is

(1) $\sqrt{\frac{3(n-1)}{(n+1)}}.100$

(2) $\sqrt{\frac{(n-1)}{3(n+1)}}.100$

(3) $\sqrt{\frac{(n+1)}{3(n-1)}}.100$

(4) None of these

69. Let $0<\theta<\frac{\pi}{4}$. Then

(1) $\sum_{k=0}^{\infty}\sin^{2k}\theta=\sec^2\theta$

(2) $\sum_{k=0}^{\infty}\cos^{2k}\theta=\tan^2\theta$

(3) $\sum_{k=0}^{\infty}\tan^{2k}\theta=\frac{\sin^2\theta}{\cos 2\theta}$

(4) $\sum_{k=0}^{\infty}\tan^{2k}\theta+\sum_{k=0}^{\infty}\cot^{2k}\theta=1$

70. Let $<a_n>$ is an infinite geometric sequence with first term 2 cot x and common ratio $\sin^2 x$ and $<b_n>$ is an infinite geometric sequence with first term $\sin 2x$ and common ratio $\sin^2 x$. If $x\in\left(0,\frac{\pi}{4}\right]$, then the minimum value of $\sum_{i=1}^{\infty}a_i-\sum_{j=1}^{\infty}b_j$ is

(1) 3 (2) 5

(3) 1 (4) 2

Integer Type Questions

71. Let $z = 1 + i$ and $z_1=\frac{1+i\,\bar{z}}{\bar{z}(1-z)+\frac{1}{z}}$.

Then value of $\frac{20\left(\text{argument}\left(z_1\right)\right)}{\pi}$ is equal to

72. If $\frac{(x-3)^{\frac{-|x|}{x}}\sqrt{(x-4)^2}\,(17-x)}{\sqrt{-x}\left(-x^2+x-1\right)\left(|x|-32\right)}<0$ then no. of integers x satisfying the inequality is:

73. Out of six faces, die A has 3 red and 2 white faces, whereas die B has 2 red and 3 white faces. A coin is tossed at random. If it shows a head, the game continues by throwing the die A and if it shows tail, then the die B is to be used. If the probability that every time die B is used in the first n throws is $\frac{16}{97}$, when it is given that every time red turns up in first n throws, then the value of n is

74. If $A=\lim_{x\to 0}\frac{\sin x+ae^x+be^{-x}+c\ln(1+x)}{x^3}$, where A is a finite non-zero number, then greatest integer value of y which satisfy in equation $[|y + a| – b] \le 2c$ is, where [.] represents the greatest integer function.

75. Let $f : R \to R$ and $g : R \to R$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define $h : R \to R$ by $h(x)=\begin{cases}\max & \{f(x),g(x)\} & \text{if} & x\le 0\\ \min & \{f(x),g(x)\} & \text{if} & x>0\end{cases}$

The number of points at which $h(x)$ is not differentiable is



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if